PURDUE UNIVERSITY | Lyles School of Civil and Construction Engineering

CE 270 | Introductory Structural Mechanics (Spring 2026)

Course Information

- **Course number and title: CE 270 | Introductory Structural Mechanics**
- **Meeting days and times:**

Section	CRN	Type	Day(s), Time, and Location
007	12667	Lecture	MWF, 08:30 AM – 09:20 AM, FRNY G140
012	12325	Lab Prep	Tue, 07:30 AM – 08:20 AM, Univ. Church 114
001	12661	Lab	Tue, 09:30 AM – 11:20 AM, HAMP G157
003	12663	Lab	Tue, 11:30 AM – 01:20 PM, HAMP G157
008	41369	Lab	Tue, 01:30 PM – 03:20 PM, HAMP G157
009	41396	Lab	Tue, 03:30 PM – 05:20 PM, HAMP G157
010	41397	Lab	Wed, 01:30 PM – 03:20 PM, HAMP G157
013	23149	Lab	Thu, 09:30 AM – 11:20 AM, HAMP G157
014	16548	Lab	Thu, 11:30 AM – 01:20 PM, HAMP G157
015	26757	Lab	Thu, 01:30 PM – 03:20 PM, HAMP G157
016	36156	Lab	Thu, 03:30 PM – 05:20 PM, HAMP G157

- **Instructional Modality: Face-to-face**

- **Course credit hours: 4**
  Lectures: 3 times per week for 50 minutes per meeting for 15 weeks
  Lab Prep: One 50-minute lab prep meeting per week for 15 weeks
  Labs: One 2-hour lab per week for 15 weeks

- **Prerequisites:**
  CE 29700 Basic Mechanics I (Statics) – Minimum Grade C-
  PHYS 17200 Modern Mechanics – Minimum Grade D-
  MA 26100 Multivariate Calculus – Minimum Grade C-
  Course status: Required core course in Civil Engineering curriculum.

Instructor and TAs Contact Information

Section	Type	Instructor or TA	Email	Phone	Office hours	Office Location
007	Lecture	Dr. Soleimani	sms110@purdue.edu	(765) 496-0157	Wed 2:30PM-5:00PM	HAMP 4127
003	Lab	Kai Smith	smit6700@purdue.edu	-	Fri 2:00PM-4:00PM	HAMP 4129
014	Lab	Kai Smith				
008	Lab	Elizabeth Geiser	egeiser@purdue.edu	-	Wed 11:20AM-1:20PM	
010	Lab	Elizabeth Geiser				
009	Lab	Juan Torres	torrestu@purdue.edu	-	Fri 4:00PM-6:00PM	
016	Lab	Juan Torres				
001	Lab	Bhoshaga Chandran	bravicha@purdue.edu	-	Thu 3:30PM-5:30PM	
015	Lab	Bhoshaga Chandran				
013	Lab	Mehran Sahil	msahil@purdue.edu	-	Mon 12:00PM-1:00PM	

Course Description

Catalog description: Loads; structural forms; analysis of axially loaded members, flexural members, torsional members; combined loading conditions; buckling. Basic behavioral characteristics of structural elements and systems illustrated by laboratory experiments.

Learning Resources, Technology & Texts

Required:
Mechanics of Materials, 11th edition
Published by Pearson (June 15, 2022) © 2023
Author: Russell C. Hibbeler

Mastering Engineering with Pearson eText for Mechanics of Materials
ISBN-13: 9780137605279 | Published 2022

Link:
See the first announcement in Brightspace.

Suggested:
Vector Mechanics for Engineers: Statics, 2024 Release
ISBN10: 1266706526 | ISBN13: 9781266706523
By Ferdinand Beer, E. Johnston and David Mazurek
https://www.mheducation.com/highered/product/vector-mechanics-engineers-statics-2024-release-beer-johnston/M9781266706523.html

Lab Safety (Required): Safety Glasses + Closed toe shoes

Purdue Libraries and School of Information Studies

- I encourage you to visit Ask a Librarian to connect with helpful resources and services provided by the Purdue Libraries and School of Information Studies for course assignments and projects.

Software/web resources

- Remember that MS Office is free for all students.

Academic Support

- The Helen Bass Williams Academic Success Center, provides a variety of proactive, practical and approachable academic support services for you to strengthen your approaches and strategies for learning, including study skills consultations, peer coaching, workshops, and online handouts. Visit the ASC website for more information and to access resources.

Brightspace learning management system (LMS)

- Access the course via Purdue's Brightspace learning management system. Begin with the Start Here tab, which offers further insight into the course and how you can be successful in it. It is strongly suggested that you explore and become familiar not only with the site navigation, but also with the content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Learning Outcomes

Specific Goals for the course:

- **Student learning outcomes** - Upon successful completion of this course the student shall be able to:
  - Analyze how different types of structural components carry loads i.e. members with axial loads, members that bend, members with torsion, and combinations.
  - Determine stresses and strains in these types of structural components.
  - Conduct experiments to evaluate the theoretically modeled behavior of these members
- **Relationship of course to program outcomes**
  - Outcome 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
  - Outcome 6: An ability to develop and conduct appropriate experimentation, analyze and interpret the data, and use engineering judgement to draw conclusions

Assignments

Tentative Schedule of Due of Dates

Week	Date (Monday)	Topic	ME/HW Due Saturday: Mastering Eng (ME) Sunday: Homework (HW)	Lab Problem Set / Lab Report Lab Report Due on Friday
1	Jan 12		HW-1 and ME-1	Problem Set #1
2	Jan 19		HW-2 and ME-2	
3	Jan 26		HW-3 and ME-3	Problem Set #2 + Report 1
4	Feb 02		HW-4 and ME-4	
5	Feb 09	Exam-1	Exam on 02/10/2026	Problem Set #3 + Report 2

6	Feb 16		HW-5 and ME-5	
7	Feb 23		HW-6 and ME-6	Problem Set #4 + Report 3
8	Mar 02		HW-7 and ME-7	Problem Set #5
9	Mar 09		HW-8 and ME-8	
10	Mar 16	Spring Break		
11	Mar 23		HW-9 and ME-9	Problem Set #6 + Report 4
12	Mar 30		HW-10 and ME-10	
13	Apr 06	Exam-2	Exam on 04/08/2026	Report 5
14	Apr 13		HW-11 and ME-11	Report 6
15	Apr 20		HW-12 and ME-12	Report 7 + Experiment 8 Handout
16	Apr 27	Exam-3	Exam on 04/28/2026	
17	May 04	**No Final Exam**		

* Schedule and assignments subject to change. Any changes will be posted in the learning management system.

Grading Scale

Grading Basis:

- Homework Assignments 25%
- Laboratories & Problem Sets 25%
- Exam 1 15% - Tue, **Feb-10**, 8:00pm-9:00pm, Location: MTHW 210 (capacity 220), and MATH 175 (capacity 180)
- Exam 2 15% - Wed, **Apr-08**, 8:00pm-9:00pm, Location: BHEE 129 (capacity 468)
- Exam 3 20% - Tue, **April-28**, 8:00pm-9:30pm, Location: WTHR 200 (capacity 480)

Total **100%**

- Bonus: 5% - Active participation in class discussions at the sole discretion of the instructor

All exams are closed book and closed notes. However, students will be provided with equation sheet(s) during the exams. Bring ONLY the following items to your exams: Black and/or blue pens, a calculator, and a ruler.

Note: Students must notify the instructor about conflicts with the scheduled exams, allowing sufficient time to verify the conflict and arrange an alternate time. Make-up exams for absences will **NOT** be given except under extremely unavoidable situations.

You may use the following grading scheme as a rough guideline to estimate your final grade in the course.

A+ / A / A- : 85 – 100
B+ / B / B- : 75 – 90
C+ / C / C- : 65 – 80
D+ / D / D- : 55 – 70
F : 0 – 60

Note: The grade cutoffs are overlapping, and they are only estimates and may be adjusted up or down at the sole discretion of the instructor at the end of the semester.
See the Spring 2026 Add/Drop calendar and the 2025-2026 Academic calendar for deadlines.

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled.

- Attendance is required in **all** lectures, labs and lab-prep sessions.
- **Cell phone** use for talking or texting is **not permitted**.
- **Avoid** arriving late, leaving early, leaving & reentering the class during the lecture.
- Ask questions:
  - **How** does this work?
  - **Why** is this so?
  - Could you please explain again …

Course Schedule

List of Topics:

1. Review of Statics:
   Free-body diagrams, Equilibrium; Distributed forces; Moments of area
2. Stress, Strain and Material Properties
   a. Stresses (Hibbeler: Ch-1)
   b. Strains (Hibbeler: Ch-2)
   c. Stress-strain relations (Hibbeler Ch-3)
   d. Stress transformation; Principal stresses; Mohr's circle (Hibbeler Ch-9)
   e. Strain transformation (Hibbeler Ch-10)
3. Axial loads
   a. Deformation (Hibbeler: Secs. 4.1-4.5)
   b. Thermal stress (Hibbeler: Sec. 4.6)
   c. Stress concentrations (Hibbeler: Sec. 4.7)
4. Bending
   a. Shear and bending moment diagrams (Hibbeler: Sec. 6.1)
   b. Load, shear, and bending moment relations (Hibbeler: Sec. 6.2)
   c. Deformation and stresses (Hibbeler: Secs. 6.3-6.4)
   d. Composite beams (Hibbeler: Secs. 6.6-6.7)
5. Transverse shear
   a. Shear stresses (Hibbeler: Secs. 7.1-7.2)
   b. Shear flow (Hibbeler: Secs. 7.3-7.4)
6. Deflections of beams and shafts
   a. The elastic curve (Hibbeler: Secs. 12.1)
   b. Slope and displacement (Hibbeler: Secs. 12.2,12.5)
   c. Statically indeterminate beams (Hibbeler: Secs. 12.6-12.7, 12.9)
7. Torsion
   a. Elastic stresses and strains (Hibbeler: Secs. 5.1-5.2)
   b. Elastic deformation (Hibbeler: Secs. 5.4-5.5)
8. Combined loadings
   a. Pressure vessels (Hibbeler: Sec. 8.1)
   b. Combined loading (Hibbeler: Sec. 8.2)
   c. Unsymmetric bending (Hibbeler: Sec. 6.5)
9. Buckling of columns
   Euler's buckling formula; Various types of supports (Hibbeler: Sec. 13.1 - 13.3)

Topics to be covered in the Labs:

1. Truss analysis
    a. Method of joints (Beer et al.: Sec. 6.1)
    b. Method of sections (Beer et al.: Sec. 6.2)
2. Cable systems
    a. Cables with concentrated loads (Beer et al.: Sec. 7.4A)
    b. Cables with distributed loads (Beer et al.: Sec. 7.4B)
    c. Parabolic cables (Beer et al.: Sec. 7.4C)
    d. Catenary (Beer et al.: Sec. 7.5)

Lab Schedule

Lab Prep: Tue: 07:30 AM – 08:20 AM, University Church 114.
Attendance **required** to conduct Lab sessions
Labs: Various times on Tue, Wed, Thu HAMP-G157 – see schedule on Page 1

Week #	**Dates**	Activity
1	Jan 13-15 T-W-R	Problem set #1 – Cables Concentrated Loads
2	Jan 20-22 T-W-R	**Experiment #1**: Cables
3	Jan 27-29 T-W-R	Problem set #2 – Cables Distributed Loads
4	Feb 03-05 T-W-R	**Experiment #2**: Axial Tension Member
5	Feb 10-12 T-W-R	Problem set #3 – Trusses
6	Feb 17-19 T-W-R	**Experiment #3**: Trusses
7	Feb 24-26 T-W-R	Problem set #4 – Beam SFD-BMD / Plane Stress
8	Mar 04-06 T-W-R	Problem set #5 – Beams stresses
9	Mar 10-12 T-W-R	**Experiment #4**: Beam stresses & strains
10	Mar 16-20 MTWRFS	Spring Break
11	Mar 24-26 T-W-R	Problem set #6 – Beams shear + deflections
12	Mar 31-Apr 02 T-W-R	**Experiment #5**: Beam deflections
13	Apr 07-09 T-W-R	**Experiment #6**: Torsion
14	Apr 14-16 T-W-R	**Experiment #7**: Combined Torsion and Bending
15	Apr 21-23 T-W-R	**Experiment #8**: Column Buckling
16	Apr 28-30 T-W-R	Optional Problem set #7 – Final Exam Review

* Schedule and assignments subject to change. Any changes will be posted in the learning management system.

Artificial Intelligence (AI) Policy

The use of generative AI tools (such as ChatGPT, Copilot, etc.) is strictly prohibited in this course for any stage of assignment creation (brainstorming, drafting, or editing). **Use of these tools will be considered a violation of the Academic Integrity policy.**
Never enter personally identifiable information (PII) or sensitive student data into public AI tools. Your interactions with these tools may be saved and used to train future models.

Course materials are undergoing remediation per the ADA Title II Ruling. Efforts are being made to the best of my ability to ensure that course materials are ADA compliant; however, these materials have not been vetted by a third party. If you are having difficulties accessing materials, please contact me immediately for support. Any materials needing further remediation will be completed by the beginning of Fall 2026.

Academic Integrity

- Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace under University Policies and Statements.
- All work (assignments and exams) that you submit must be strictly **your own work**.
- Obtaining solutions from an external source or from another student's homework and/or sharing your homework with another student is **absolutely not allowed**. Giving and receiving help on **concepts** is allowed and encouraged.

Take the Purdue Honor Pledge: "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under *University Policies*.

Accessibility

Purdue University is committed to making learning experiences accessible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document (**PDF**) or (**Word**).*

The first day of class, I will review the **Emergency Preparedness plan for our specific classroom**, following Purdue's required Emergency Preparedness Briefing. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.